Data Structures - List, Dictionary, Set

& Conditional Statements

List (Creation, Modification and Access):
1. List Creation:
a. Create a list named age_list with five integer elements. For eg., [24, 25, 26, 27, 28]
b. Create a list named name_list with five string elements.
2. List Operations / Modifications:
a. Append the string "Yazhini" to name_list.
b. Insert the element 30 at index 2 in age_list.
c. Remove the string "Yazhini" from name_list.
d. Pop the last element from age_list.
e. Extend the age_list with additional ages [29, 30, 26].
f. Sort age_list in descending order.
g. Find Max age, Min age and sum of all ages from age_list.
3. Accessing List Elements:
a. Print the first element of name_list.
b. Print the last element of name_list.
c. Print the elements from index 2 to index 4 in name_list.
d. Print the elements of name_list in reverse order.

Dictionary (Creation, Modification and Access):
a. Create a dictionary named student_marks that maps the names of five students to their marks (use scale of from 0 to 100).
b. Access and print the mark of a specific student, of your choice.
c. Add a new student "Janani" with a mark of 80 to the student_marks dictionary.
d. Update the mark of any one older student to 82.

e. Use the keys(), values(), and items() methods to print all keys, values, and key-value pairs in the student_marks dictionary.

Sets (Operations):
a. Create a set called my_set with following values:
['a','e','i','o','u','a','a','i']
Analyse the output and provide explanation for the same.
b. Attempt to change the value of my_set[4] = 's'. If code throws an error, provide an explanation.
c. Create two sets:

set1 with values: {1, 3, 5, 7, 9}
set2 with values: {2, 3, 5, 8, 10}

d. Compute and print the union and intersection of set1 and set2.

Operators & Conditional Statements :
(IF, ELIF, ELSE)
Performance Category Program:
1. Prompt user for Input. Score range should be from 0 to 10 (both inclusive).
2. Find the performance category based on the input score using following criteria:
a. Above Average: Score greater than 7
b. Average: Score between 4 and 7(both inclusive)
c. Below Average: Score lesser than 4
3. Output: Print the Performance category
4. Additional Step: You can give a prompt of your choice to each category.
For eg: If score below average “Need to Improve your performance, consistent practice will lead to better results”.

Sample Output:
Enter your score (0 to 10): 7
Average: Good effort! Keep practicing, there's room for improvement.